Implementation of Shrinkage Estimators for Cosmological Precision Matrices in TEDA: Enhancing Ensemble-Based Data Assimilation in High-Dimensional Settings*

Elias D. Nino-Ruiz[1][0000-0001-7420-4517] and Andrés Movilla Obregón[1]

Department of Computer Science and Statistics, Universidad del Norte, Barranquilla, Colombia
{enino, movillaf}@uninorte.edu.co

Abstract. Ensemble Kalman Filters require accurate precision matrix estimates, yet the sample covariance becomes ill-conditioned when the ensemble size is comparable to the state dimension. We present the implementation of six shrinkage estimators within the TEDA (Toolbox for Ensemble Data Assimilation) educational framework: three direct precision shrinkage variants from Looijmans et al. (identity, eigenvalue, and cosmological targets), a novel scaled identity adaptation, the NERCOME estimator, and Ledoit-Wolf covariance shrinkage. Dual-domain experiments on the Lorenz 96 system and BOSS DR12 cosmological mock data reveal that direct precision shrinkage performance depends critically on target-truth alignment: these methods fail on Lorenz 96 (dense off-diagonal precision) but achieve excellent standalone estimation on near-diagonal cosmological data. Surprisingly, standalone estimation quality does not predict filtering performance—Ledoit-Wolf yields the worst precision matrix estimates on cosmological data yet the best EnKF RMSE, suggesting that estimator stability across sequential assimilation cycles matters as much as per-step accuracy. All implementations are publicly available in TEDA, with code structure mirroring the mathematical exposition to support teaching and reproducible research.

Keywords: Data Assimilation · Ensemble Kalman Filter · Shrinkage Estimation · Cosmological Precision Matrix · Covariance Matrix Estimation · TEDA · Python

1 Introduction

Ensemble-based data assimilation has become a cornerstone methodology in numerical weather prediction, oceanography, and increasingly in cosmological parameter estimation [5,2]. The success of ensemble methods fundamentally depends on accurate estimation of background error covariances, which becomes

* TEDA source code is available at: https://github.com/enino84/TEDA.git

particularly challenging in high-dimensional systems where the ensemble size N_e is limited relative to the state dimension n. In operational settings, state dimensions can reach millions while ensemble sizes are typically bounded by the hundreds, making the sample covariance matrix rank-deficient and unsuitable for direct inversion [10].

The TEDA (Toolbox for Ensemble Data Assimilation) framework [16] was developed as an educational platform to facilitate the teaching and learning of ensemble-based data assimilation methods. Written in Python and following the Object-Oriented Programming (OOP) paradigm, TEDA provides implementations of several well-known ensemble methods alongside toy dynamical models such as the Lorenz 96 [14] and quasi-geostrophic systems. While originally designed for meteorological applications, the growing importance of data assimilation in cosmological applications has motivated the extension of TEDA to address cosmological precision matrix estimation challenges.

In cosmological data assimilation, observations typically consist of power spectrum measurements at various multipole moments ℓ and redshift bins, resulting in data vectors whose covariance structure is intimately connected to the underlying cosmological power spectrum C_ℓ [8]. The covariance matrices arising in this context have a well-understood theoretical structure governed by cosmic variance and shot noise contributions. Traditional sample covariance estimators perform poorly in this regime due to the curse of dimensionality: when $N_e < n$, the sample covariance matrix $\mathbf{P}^b$ becomes singular, making direct computation of the precision matrix $(\mathbf{P}^b)^{-1}$ impossible. Even when N_e is moderately larger than n, sampling noise introduces spurious correlations that degrade the quality of the estimated precision matrix [1].

Shrinkage estimation provides a principled approach to regularize covariance matrix estimation by combining the sample covariance with a well-conditioned target matrix [12]. The key insight is that while the sample covariance is unbiased but high-variance, a structured target matrix is biased but low-variance; the optimal combination of both can achieve lower mean squared error than either alone. Recent advances by Pope and Szapudi [17] and Looijmans et al. [13] have developed specialized shrinkage estimators tailored to cosmological applications, leveraging the known structure of cosmological covariance matrices to improve estimation accuracy. These methods are particularly relevant because the cosmological covariance matrix has a predictable diagonal structure related to the power spectrum, providing a natural and informative shrinkage target.

This paper presents a comprehensive implementation of these advanced shrinkage estimators within the TEDA framework, making these sophisticated methods accessible to the data assimilation and cosmology communities. Our contributions include:

1. Implementation of six shrinkage estimators within TEDA: three direct precision shrinkage variants from Looijmans et al. [13] (identity, eigenvalue-based, and cosmological targets), one novel domain-agnostic adaptation proposed in this work (scaled identity target), the NERCOME estimator of Joachimi [11]

as applied in Looijmans et al., and the Ledoit-Wolf covariance shrinkage estimator [12].
2. Integration of all six methods into the TEDA object-oriented architecture, maintaining the framework's educational design philosophy while adding research-grade capabilities.
3. Dual-domain experimental validation using both the Lorenz 96 chaotic dynamical system and BOSS DR12 cosmological mock catalog data [13], demonstrating that direct precision shrinkage performance depends critically on target-truth alignment.
4. Discovery that standalone precision matrix estimation quality does not directly predict filtering performance: Ledoit-Wolf achieves the worst standalone estimation on cosmological data but the best RMSE within the EnKF pipeline, highlighting the importance of estimator stability in sequential data assimilation.

The remainder of this paper is organized as follows: Sect. 2 provides theoretical background on ensemble-based data assimilation, shrinkage estimation, and cosmological covariance matrices. Sect. 3 details our implementation approach and the specific algorithms incorporated into TEDA. Sect. 4 presents experimental validation and performance comparisons. Sect. 5 discusses the implications for data assimilation practice and education. Finally, Sect. 6 summarizes our contributions and outlines future work.

2 Background and Theory

This section reviews the theoretical foundations underlying our implementation: ensemble-based data assimilation, shrinkage estimation theory, and the structure of cosmological covariance matrices.

2.1 Ensemble-Based Data Assimilation

The Ensemble Kalman Filter (EnKF) [5] represents the state of a dynamical system using an ensemble of N_e state vectors $\{\mathbf{x}_i^b\}_{i=1}^{N_e}$, where each member $\mathbf{x}_i^b \in \mathbb{R}^{n\times 1}$ is a realization of the forecast (background) state. The ensemble mean provides the best estimate of the background state:

$$\bar{\mathbf{x}}^b = \frac{1}{N_e}\sum_{i=1}^{N_e}\mathbf{x}_i^b \in \mathbb{R}^{n\times 1}, \tag{1}$$

and the background error covariance matrix is estimated from the ensemble perturbations. Defining the deviation matrix as:

$$\Delta\mathbf{X}^b = \mathbf{X}^b - \bar{\mathbf{x}}^b\mathbf{1}^T \in \mathbb{R}^{n\times N_e}, \tag{2}$$

where $\mathbf{X}^b = [\mathbf{x}_1^b, \ldots, \mathbf{x}_{N_e}^b]$ is the ensemble matrix, the sample background error covariance is:

$$\mathbf{P}^b = \frac{1}{N_e - 1}\Delta\mathbf{X}^b(\Delta\mathbf{X}^b)^T \in \mathbb{R}^{n\times n}. \tag{3}$$

In the analysis step, observations $\mathbf{y} \in \mathbb{R}^{m\times 1}$ are assimilated to correct the background state. The stochastic EnKF computes analysis updates by solving the linear system:

$$[\mathbf{H}\mathbf{P}^b\mathbf{H}^T + \mathbf{R}]\, \Delta\mathbf{X}^a = \mathbf{D}^s \in \mathbb{R}^{m\times N_e}, \tag{4}$$

where $\mathbf{H} \in \mathbb{R}^{m\times n}$ is the observation operator, $\mathbf{R} \in \mathbb{R}^{m\times m}$ is the observation error covariance, and $\mathbf{D}^s$ is the innovation matrix containing perturbed observation departures.

An alternative formulation works in ensemble space via the dual EnKF [15]:

$$[(\mathbf{P}^b)^{-1} + \mathbf{H}^T\mathbf{R}^{-1}\mathbf{H}]\, \Delta\mathbf{X}^a = \mathbf{H}^T\mathbf{R}^{-1}\mathbf{D}^s \in \mathbb{R}^{n\times N_e}. \tag{5}$$

This dual formulation requires the precision matrix $(\mathbf{P}^b)^{-1}$, which is the central object of interest in this work. When $N_e < n$, the sample covariance (3) is rank-deficient with rank at most $N_e - 1$, making direct inversion impossible. Even when $N_e > n$, the sample precision matrix $(\mathbf{P}^b)^{-1}$ can be poorly conditioned, amplifying sampling noise in the analysis step.

Yet another implementation projects the analysis onto the ensemble subspace [16]:

$$\mathbf{X}^a = \mathbf{X}^b + \Delta\mathbf{X}^b\mathbf{W}^*, \tag{6}$$

where the weight matrix $\mathbf{W}^* \in \mathbb{R}^{N_e\times N_e}$ is obtained by solving the reduced system in ensemble space. This formulation is computationally efficient when $N_e \ll n$, as it reduces the problem from n-dimensional to N_e-dimensional.

2.2 Shrinkage Estimation

Shrinkage estimation addresses the ill-conditioning of the sample covariance by combining it with a well-conditioned target matrix $\mathbf{T}$:

$$\hat{\mathbf{P}} = (1-\lambda)\mathbf{S} + \lambda\mathbf{T}, \tag{7}$$

where $\mathbf{S}$ is the sample covariance matrix and $\lambda \in [0,1]$ is the shrinkage intensity. When $\lambda = 0$, we recover the sample covariance; when $\lambda = 1$, we use only the target. The optimal λ minimizes the expected loss under the Frobenius norm:

$$\lambda^* = \arg\min_{\lambda\in[0,1]} \mathbb{E}\left[\|\hat{\mathbf{P}} - \boldsymbol{\Sigma}\|_F^2\right], \tag{8}$$

where $\boldsymbol{\Sigma}$ is the true (unknown) covariance matrix.

Ledoit-Wolf estimator. Ledoit and Wolf [12] derived an analytical solution for the optimal shrinkage intensity when the target is a scaled identity matrix $\mathbf{T} = \mu\mathbf{I}$, where $\mu = \mathrm{tr}(\mathbf{S})/n$:

$$\lambda^*_{\mathrm{LW}} = \frac{\sum_{i=1}^{N_e} \|\mathbf{z}_i\mathbf{z}_i^T - \mathbf{S}\|_F^2/N_e^2}{\|\mathbf{S} - \mu\mathbf{I}\|_F^2}, \tag{9}$$

where $\mathbf{z}_i$ are the standardized observations. This estimator is asymptotically optimal and has become a standard reference in high-dimensional covariance estimation.

Oracle estimator. In simulation studies where the true covariance $\boldsymbol{\Sigma}$ is known, the oracle shrinkage intensity provides a best-case benchmark:

$$\lambda^*_{\text{oracle}} = \frac{\text{tr}[(\mathbf{S} - \boldsymbol{\Sigma})(\mathbf{T} - \boldsymbol{\Sigma})]}{\|\mathbf{T} - \mathbf{S}\|_F^2}. \tag{10}$$

While not available in practice, the oracle estimator is useful for evaluating how close a data-driven estimator comes to the theoretical optimum.

Nercome estimator. The Nercome (Non-Equal Rows Covariance Matrix Estimator) [11] extends the Ledoit-Wolf framework by allowing different shrinkage intensities for different blocks of the covariance matrix, which is particularly relevant when the data vector combines measurements of different types or at different scales.

2.3 Cosmological Covariance Matrix Structure

In cosmological surveys, the data vector typically consists of power spectrum measurements $\hat{C}_\ell$ organized by multipole moments ℓ and, in tomographic analyses, by redshift bins z. The theoretical covariance matrix for such measurements has a well-understood structure based on cosmic variance and shot noise contributions [8].

For a Gaussian random field observed over a fraction f_{sky} of the sky, the covariance of power spectrum estimates at different multipoles is [17]:

$$\text{Cov}(\hat{C}_{\ell_i}, \hat{C}_{\ell_j}) = \frac{2}{(2\ell_i + 1) f_{\text{sky}}} \left(C_{\ell_i} + \frac{1}{n_g} \right)^2 \delta_{\ell_i \ell_j}, \tag{11}$$

where C_ℓ is the true power spectrum, n_g is the galaxy number density, and $\delta_{\ell_i \ell_j}$ is the Kronecker delta. The term $1/n_g$ represents shot noise, and the factor $(2\ell + 1) f_{\text{sky}}$ counts the number of independent modes N_ℓ available at multipole ℓ. The diagonal structure of (11) arises from the statistical independence of different multipole modes for a Gaussian field.

The corresponding precision matrix takes the form:

$$\mathbf{P}^{-1}_{ij} = \frac{(2\ell_i + 1)}{2} f_{\text{sky}} \left(C_{\ell_i} + \frac{1}{n_g} \right)^{-2} \delta_{\ell_i \ell_j} \, \delta_{z_i z_j}. \tag{12}$$

This known structure motivates a cosmological shrinkage target. Following Looijmans et al. [13], we define the diagonal target matrix:

$$T^{(2)}_{ii} = \frac{2}{N_{\ell_i}} C^2_{\ell_i}, \tag{13}$$

where $N_{\ell_i} = (2\ell_i + 1) f_{\text{sky}}$ is the number of modes at multipole ℓ_i. This target captures the leading-order (cosmic variance) contribution to the covariance,

providing an informative prior for shrinkage estimation. In contrast to a generic identity target, this cosmological target incorporates domain knowledge about the expected covariance structure, which can significantly improve estimation accuracy when the sample size is limited.

2.4 Related Shrinkage Approaches

Several shrinkage estimators have been proposed for cosmological precision matrices. Pope and Szapudi [17] introduced shrinkage estimation to the cosmological context, demonstrating that even a simple identity-scaled target significantly improves power spectrum covariance estimation compared to the raw sample covariance. Their approach applies shrinkage in eigenvalue space, correcting the bias in sample eigenvalues that arises from limited sample sizes.

Looijmans et al. [13] conducted a systematic comparison of multiple shrinkage estimators for cosmological precision matrices, including identity-scaled targets, diagonal cosmological targets, and eigenvalue shrinkage methods. Their analysis showed that estimators incorporating cosmological prior information (i.e., the power spectrum structure) generally outperform generic estimators, with the magnitude of improvement depending on the ratio N_e/n and the accuracy of the assumed cosmological model.

In the broader statistical literature, Haff [7] established the theoretical foundations for shrinkage estimation of precision matrices, while Chen et al. [4] developed shrinkage estimators specifically for the covariance matrix in high-dimensional settings with applications to portfolio optimization. These methods share the common principle of trading bias for variance reduction, with the target matrix encoding prior structural assumptions about the true covariance.

3 Implementation in TEDA

This section describes the integration of shrinkage-based precision matrix estimators into the TEDA framework, including the software architecture, the specific algorithms implemented, and the design decisions that enable educational use.

3.1 TEDA Architecture Overview

TEDA follows an object-oriented design pattern where dynamical models, ensemble management, observation handling, and analysis methods are implemented as modular classes. Figure 1 shows the class hierarchy relevant to our implementation.

The core classes are:

- **Model**: Abstract base class for dynamical systems (Lorenz 96, quasi-geostrophic). Defines the `propagate()` interface for time integration.
- **Background**: Manages the ensemble, including generation, propagation, and computation of ensemble statistics (mean, covariance, deviations).

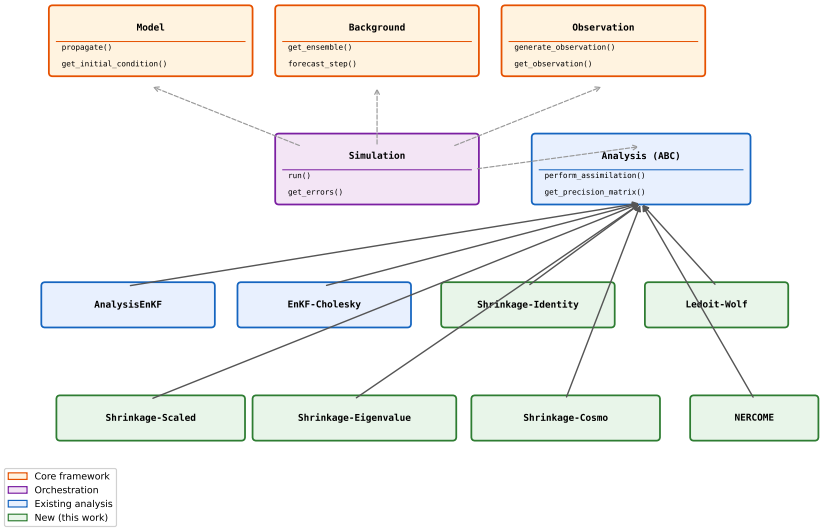

Fig. 1. Class diagram of the TEDA framework, highlighting the new shrinkage estimator classes (shaded). All analysis methods inherit from the abstract `Analysis` base class and are registered via the `AnalysisFactory`, enabling seamless interchangeability.

- **Observation**: Generates synthetic observations and manages the observation operator $\mathbf{H}$ and error covariance $\mathbf{R}$.
- **Analysis**: Abstract base class defining the assimilation interface. Subclasses implement specific methods by overriding `perform_assimilation()`.
- **Simulation**: Orchestrates the forecast-analysis cycle, computes error metrics, and stores results.

New analysis methods are registered via a factory pattern. Using the decorator `@register_analysis`, users can select methods by name at runtime, facilitating comparative studies:

```
from pyteda.analysis import AnalysisFactory
method = AnalysisFactory.create("EnKF-Shrinkage-Cosmo",
                                 params=config)
```

3.2 Shrinkage Estimator Implementations

We have implemented four variants of the *direct linear shrinkage estimator for the precision matrix*, three from Looijmans et al. [13] and one novel domain-agnostic adaptation proposed in this work. Each variant corresponds to a different choice of target precision matrix $\boldsymbol{\Pi}_0$. All implementations inherit from the `Analysis` base class and override the `get_precision_matrix()` method.

Algorithm 1 Direct Precision Shrinkage (Identity Target)

Require: Deviation matrix $\mathbf{DX} \in \mathbb{R}^{n \times N_e}$
1: Compute sample covariance: $\mathbf{S} = \frac{1}{N_e - 1} \mathbf{DX}\,\mathbf{DX}^T$
2: Compute sample precision: $\hat{\boldsymbol{\Pi}}_S = \mathbf{S}^{-1}$
3: Set target: $\boldsymbol{\Pi}_0 = \mathbf{I}_n$
4: Compute α^*, β^* via Eqs. (15)–(16)
5: **return** $\boldsymbol{\Pi}_{\mathrm{LS}} = \alpha^* \hat{\boldsymbol{\Pi}}_S + \beta^* \boldsymbol{\Pi}_0$

Following Bodnar et al. [3] as applied in Looijmans et al. [13], the direct precision shrinkage estimator combines the raw sample precision matrix $\hat{\boldsymbol{\Pi}}_S = \mathbf{S}^{-1}$ with a structured target $\boldsymbol{\Pi}_0$:

$$\boldsymbol{\Pi}_{\mathrm{LS}} = \alpha^* \hat{\boldsymbol{\Pi}}_S + \beta^* \boldsymbol{\Pi}_0, \tag{14}$$

where the optimal mixing parameters α^* and β^* are estimated analytically. Crucially, the Bodnar formula is derived for the *raw* inverse $\mathbf{S}^{-1}$; the $(1-n/N_e)$ term in Eq. (15) already encodes the estimation bias. Applying the Hartlap correction $\gamma = (N_e - n - 2)/(N_e - 1)$ [9] prior to computing α^* and β^* would scale all terms nonequivalently and alter the formula in a way not derived in the original work.

$$\alpha^* = 1 - \frac{n}{N_e} - \frac{N_e^{-1} \|\hat{\boldsymbol{\Pi}}_S\|_{\mathrm{tr}}^2 \, \|\boldsymbol{\Pi}_0\|_F^2}{\|\hat{\boldsymbol{\Pi}}_S\|_F^2 \, \|\boldsymbol{\Pi}_0\|_F^2 - \left(\mathrm{tr}(\hat{\boldsymbol{\Pi}}_S \boldsymbol{\Pi}_0)\right)^2}, \tag{15}$$

$$\beta^* = \left(1 - \frac{n}{N_e} - \alpha^*\right) \frac{\mathrm{tr}(\hat{\boldsymbol{\Pi}}_S \boldsymbol{\Pi}_0)}{\|\boldsymbol{\Pi}_0\|_F^2}, \tag{16}$$

where $\|\cdot\|_{\mathrm{tr}}$ denotes the trace norm and $\|\cdot\|_F$ the Frobenius norm. Both α^* and β^* are clamped to $[0, 1]$ with $\alpha^* + \beta^* \leq 1$.

The four variants differ only in the choice of $\boldsymbol{\Pi}_0$, as detailed below.

Identity Target. This variant uses the simplest possible target: $\boldsymbol{\Pi}_0 = \mathbf{I}_n$, assuming independent, unit-variance priors for all state variables. It is implemented in class `AnalysisEnKFDirectPrecisionShrinkageIdentity`.

While the identity target makes no domain-specific assumptions, the direct precision formulation ensures that the resulting estimator always produces a valid precision matrix, unlike inverting a shrunk covariance which can amplify estimation error.

Eigenvalue Target. This variant uses the eigenvalue-based target from Looijmans et al. Eq. (15): the target precision matrix is a diagonal matrix of the sorted eigenvalues of the sample precision $\hat{\boldsymbol{\Pi}}_S$. The TEDA class is `AnalysisEnKFDirectPrecisionShrinkageEigenvalues`.

This target preserves the spectral information of the sample precision while removing off-diagonal correlations. It requires no external domain knowledge, making it applicable to general data assimilation problems. The eigenvalue target is the first precision matrix target ($\boldsymbol{\Pi}_0^{(1)}$) evaluated in Looijmans et al. [13].

Algorithm 2 Direct Precision Shrinkage (Eigenvalue Target)

Require: Deviation matrix $\mathbf{DX} \in \mathbb{R}^{n \times N_e}$
1: Compute $\mathbf{S}$, $\hat{\boldsymbol{\Pi}}_S$ as in Algorithm 1 (Steps 1–2)
2: Eigendecomposition: $\hat{\boldsymbol{\Pi}}_S = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^T$
3: Set target: $\boldsymbol{\Pi}_0^{(1)} = \mathrm{diag}(e_1, \ldots, e_n)$ where e_i are the eigenvalues of $\hat{\boldsymbol{\Pi}}_S$ in ascending order
4: Compute α^*, β^* via Eqs. (15)–(16)
5: **return** $\boldsymbol{\Pi}_{\mathrm{LS}} = \alpha^* \hat{\boldsymbol{\Pi}}_S + \beta^* \boldsymbol{\Pi}_0^{(1)}$

Algorithm 3 Direct Precision Shrinkage (Cosmological Target)

Require: Deviation matrix $\mathbf{DX}$, power spectrum C_ℓ, mode counts N_ℓ
1: Compute $\mathbf{S}$, $\hat{\boldsymbol{\Pi}}_S$ as in Algorithm 1 (Steps 1–2)
2: **for** each data element i mapped to multipole ℓ_i **do**
3: Compute diagonal covariance target: $T_{ii}^{(2)} = \frac{2}{N_{\ell_i}} C_{\ell_i}^2$
4: **end for**
5: Set target precision: $\boldsymbol{\Pi}_0^{(2)} = (\mathbf{T}^{(2)})^{-1}$
6: Compute α^*, β^* via Eqs. (15)–(16)
7: **return** $\boldsymbol{\Pi}_{\mathrm{LS}} = \alpha^* \hat{\boldsymbol{\Pi}}_S + \beta^* \boldsymbol{\Pi}_0^{(2)}$

Scaled Identity Target (novel adaptation). This novel adaptation, proposed in this work, applies the Bodnar framework without domain-specific prior knowledge. Since real power spectrum data C_ℓ and mode counts N_ℓ are unavailable outside the cosmological setting, we substitute empirical per-variable variances: $\boldsymbol{\Pi}_0 = \mathrm{diag}(1/(2\sigma_1^2), \ldots, 1/(2\sigma_n^2))$, where σ_i^2 are the diagonal entries of $\mathbf{S}$. The factor of 2 mirrors the $2/N_\ell$ scaling of the cosmological target. This method is *not* from Looijmans et al.; its experimental results (Section 4) illustrate how much the performance of direct precision shrinkage depends on domain-specific target knowledge. The TEDA class is `AnalysisEnKFDirectPrecisionShrinkageIdentityScaled`.

Cosmological Target. This variant implements the analytical cosmological target from Looijmans et al. Eqs. (14) and (16), available in TEDA as `AnalysisEnKFDirectPrecisionShrinkageCosmo`. The target precision matrix is the inverse of the theoretical Gaussian covariance of the power spectrum:

The cosmological target incorporates the known diagonal structure of power spectrum covariances, where $T_{ii}^{(2)} = (2/N_{\ell_i})C_{\ell_i}^2$ represents the Gaussian cosmic variance at multipole ℓ_i [8]. This domain knowledge provides a more informative shrinkage direction than the identity or eigenvalue targets when the assumed power spectrum model is close to the truth.

3.3 Integration and Usage

All shrinkage estimator classes are registered in the analysis factory, allowing seamless integration with existing TEDA workflows. The implementation follows TEDA's modular design principles:

- **Modularity**: Each shrinkage method is implemented as a separate class in its own file under `pyteda/analysis/`.
- **Extensibility**: New shrinkage variants can be added by subclassing `Analysis` and registering with `@register_analysis`.
- **Consistency**: All classes follow the same interface pattern, enabling direct comparison in a single simulation run.
- **Educational focus**: Code includes docstrings with mathematical notation matching this paper, and default parameters are chosen to produce meaningful results without tuning.

A typical experiment comparing multiple methods requires only changing the analysis method name while keeping all other parameters fixed:

```
methods = ["EnKF", "EnKF-Cholesky",
           "EnKF-Shrinkage-Identity",
           "EnKF-Shrinkage-Cosmo",
           "EnKF-Shrinkage-Eigenvalues"]
for name in methods:
    sim = Simulation(model, background,
                     AnalysisFactory.create(name),
                     observation)
    sim.run()
```

This design enables comparative studies with minimal code changes, making it straightforward for students to explore the effects of different precision matrix estimation strategies on assimilation quality.

4 Experimental Results

We validate our implementations using synthetic experiments with the Lorenz 96 model [14], a standard benchmark in data assimilation research. The chaotic dynamics of this model ($n = 40$ state variables, forcing parameter $F = 8$) provide a challenging test bed for covariance estimation methods.

4.1 Experimental Setup

All experiments share the following configuration unless otherwise stated:

- **Model**: Lorenz 96 with $n = 40$ variables and forcing $F = 8$.
- **Observations**: $m = 32$ randomly placed observations with Gaussian error $\sigma_o = 0.01$.

- **Ensemble sizes**: $N_e \in \{50, 60, 80, 100\}$ (all satisfying $N_e > n + 2 = 42$, required for direct precision methods).
- **Assimilation window**: 100 analysis cycles with observation frequency $\Delta t_{\text{obs}} = 0.1$ time units (total simulation time $T = 10$).
- **Inflation factor**: Multiplicative inflation with factor $\rho = 1.05$.
- **Repetitions**: Each configuration is run 10 times with different random seeds to compute mean and standard deviation of error metrics.

The seven methods compared are: standard EnKF, EnKF with modified Cholesky decomposition (bandwidth $r = 2$), three direct precision shrinkage variants (identity target, scaled identity target, eigenvalue target), the NERCOME estimator [11], and Ledoit-Wolf covariance shrinkage. Of these, the scaled identity target is a novel domain-agnostic adaptation proposed in this work (Section 3.2); the remaining methods follow Looijmans et al. [13]. The cosmological target (Algorithm 3) is not tested on Lorenz 96, as this model does not produce power spectrum data; it is evaluated on cosmological data in Section 4.7.

4.2 Error Metrics

We evaluate performance using the relative Root Mean Square Error (RMSE) of the analysis state:

$$\text{RMSE}(t) = \frac{\|\bar{\mathbf{x}}^a(t) - \mathbf{x}^{\text{true}}(t)\|_2}{\|\mathbf{x}^{\text{true}}(t)\|_2}, \tag{17}$$

where $\bar{\mathbf{x}}^a(t)$ is the analysis ensemble mean and $\mathbf{x}^{\text{true}}(t)$ is the true state at time t. We report the time-averaged RMSE over the last 50 assimilation cycles (after the initial transient).

4.3 Comparison Across Methods

Figure 2 shows the evolution of RMSE over 100 assimilation cycles for six methods with $N_e = 100$ ensemble members.

4.4 Sensitivity to Ensemble Size

Figure 3 shows the time-averaged RMSE as a function of N_e for each method. While the standard EnKF and Ledoit-Wolf show the expected monotonic improvement with larger ensembles, the direct precision shrinkage methods exhibit erratic, non-monotonic behavior: RMSE can increase substantially at certain ensemble sizes before decreasing again. This instability is analyzed in detail in Section 5.

4.5 Precision Matrix Quality

To directly assess the quality of precision matrix estimation, we compare the estimated precision matrices from different estimators at a representative assimilation step. Figure 4 shows heatmaps of the precision matrices produced by each method with $N_e = 60$.

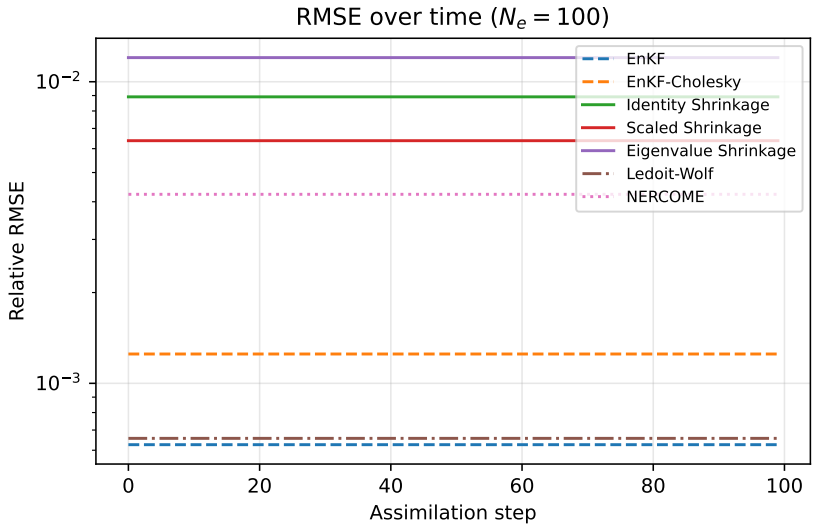

Fig. 2. Evolution of relative RMSE over 100 assimilation cycles for different analysis methods with $N_e = 100$. Shrinkage-based methods (solid lines) and Ledoit-Wolf (dash-dot) are compared against the standard EnKF and EnKF-Cholesky (dashed).

4.6 Quantitative Comparison

Table 1 summarizes the performance of all methods across different ensemble sizes. For each configuration, we report the mean ± standard deviation of the time-averaged RMSE over 10 independent runs. The direct precision shrinkage methods show high variance across runs, confirming the seed-dependent instability observed in Figure 3.

4.7 Validation with Cosmological Data

The Lorenz 96 results demonstrate that direct precision shrinkage is sensitive to target-truth mismatch. To test whether these methods perform as intended when the target *does* match the data structure, we conduct two complementary experiments using BOSS DR12 Patchy mock catalog data from Looijmans et al. [13]. The mock catalogs provide 2048 power spectrum realizations with $d = 18$ bins, whose covariance is approximately diagonal—precisely the regime for which the Bodnar et al. [3] estimators were designed.

Part A: Direct Precision Matrix Estimation. We first replicate the direct comparison setup of Looijmans et al. For each subsample size $n \in \{24, 30, 40, 50\}$, we draw $M = 100$ random subsets of n mocks, compute each estimator's precision matrix from the subsample, and measure the relative Frobenius loss against

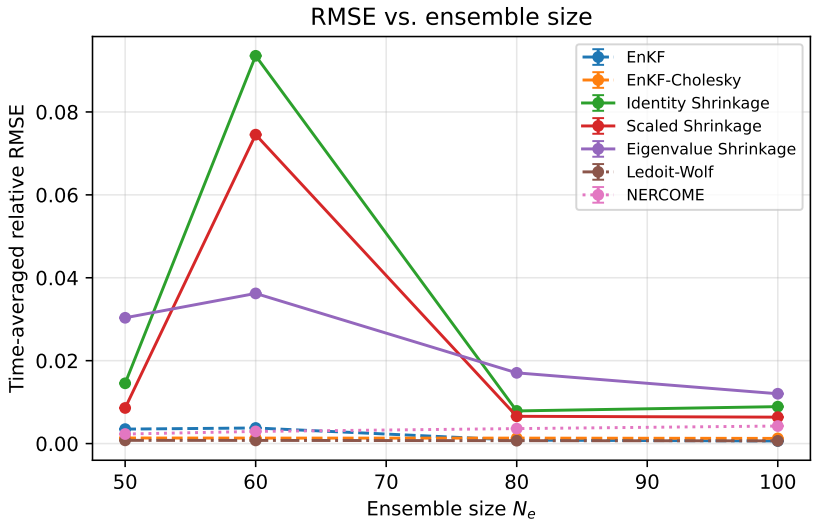

Fig. 3. Time-averaged relative RMSE (over last 50 cycles) as a function of ensemble size N_e. Error bars show ± 1 standard deviation over 10 runs. The direct precision shrinkage methods exhibit non-monotonic behavior, while Ledoit-Wolf achieves consistently low error across all ensemble sizes.

the "true" precision matrix computed from all 2048 mocks (with Hartlap correction [9]):

$$\mathcal{L}(\hat{\boldsymbol{\Pi}}, \boldsymbol{\Pi}_{\text{true}}) = \frac{\|\hat{\boldsymbol{\Pi}} - \boldsymbol{\Pi}_{\text{true}}\|_F}{\|\boldsymbol{\Pi}_{\text{true}}\|_F}. \tag{18}$$

Table 2 shows the results. In contrast to the Lorenz 96 experiment, the direct precision shrinkage methods now outperform Ledoit-Wolf by a wide margin. At $n = 50$, the identity, eigenvalue, and cosmological targets all achieve a loss of ≈ 0.41, compared to 0.93 for Ledoit-Wolf. This confirms the findings of Looijmans et al.: when the true precision matrix is near-diagonal, direct precision shrinkage with diagonal targets is highly effective.

Figure 5 visualizes these results. Notably, Ledoit-Wolf's loss remains nearly constant at ≈ 0.93–0.96 regardless of subsample size. This reflects its design: Ledoit-Wolf shrinks the *covariance* matrix toward a diagonal target, then inverts. When the goal is to estimate a near-diagonal *precision* matrix, this indirect approach is suboptimal. NERCOME performs well at small n (best at $n = 24$) due to its low variance, while the novel Scaled Shrinkage target achieves the lowest mean loss at $n = 30$. Both are overtaken by the Bodnar methods at larger sample sizes where the sample precision becomes more reliable.

Part B: EnKF with AR(1) Cosmological Model. To test these methods within the full TEDA pipeline, we construct an AR(1) dynamical model

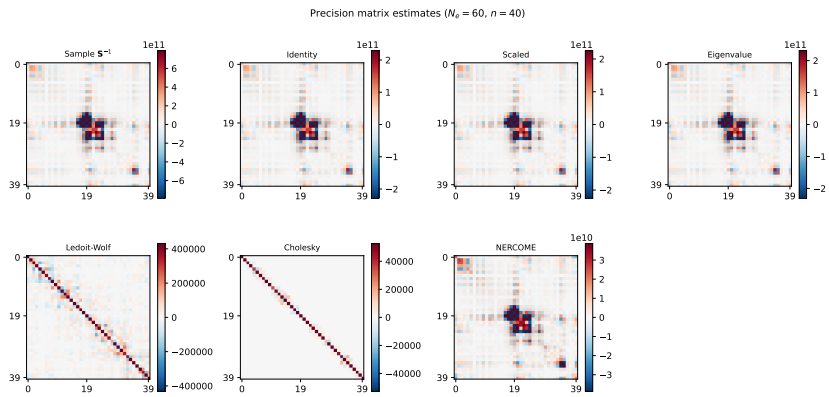

Fig. 4. Precision matrix estimates (40×40) from different methods with $N_e = 60$: sample precision $\mathbf{S}^{-1}$, identity shrinkage, scaled shrinkage, eigenvalue shrinkage, Ledoit-Wolf, and modified Cholesky. Note the difference in scale: the direct precision shrinkage estimators produce matrices with much larger magnitude than Ledoit-Wolf and Cholesky, reflecting the ill-conditioning discussed in Section 5.

calibrated to the mock catalog statistics:

$$\mathbf{x}_{t+1} = \phi\, \mathbf{x}_t + (1-\phi)\, \boldsymbol{\mu} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \tag{19}$$

where $\boldsymbol{\mu}$ is the mean power spectrum from the mock catalogs, $\phi = 0.95$ is a persistence parameter, and $\mathbf{Q} = (1-\phi^2)\boldsymbol{\Sigma}_{\text{mock}}$ is chosen so that the stationary distribution has covariance $\boldsymbol{\Sigma}_{\text{mock}}$. This preserves the near-diagonal precision structure of the cosmological data within a dynamical system compatible with TEDA's EnKF pipeline.

The experiment uses $d = 18$ state variables, $m = 14$ observations ($\approx 78\%$ coverage, comparable to 32/40 for Lorenz 96), observation error $\sigma_o = 100$, inflation factor $\rho = 1.02$, and ensemble sizes $N_e \in \{24, 30, 40, 50\}$. All eight methods are tested, including the cosmological shrinkage target configured with the BOSS power spectrum parameters. Table 3 shows the time-averaged RMSE.

Figure 6 shows the RMSE as a function of ensemble size. All methods converge toward similar performance as N_e increases, with Ledoit-Wolf maintaining a slight edge across all sizes. The standard EnKF starts with substantially higher error at small ensembles ($N_e = 24$) but improves rapidly, while the shrinkage-based methods provide more consistent performance from the start. Unlike the Lorenz 96 experiments (Figure 3), the direct precision shrinkage methods show smooth, monotonic improvement with ensemble size—confirming that the erratic Lorenz 96 behavior was due to target-truth mismatch rather than a fundamental limitation. Pairwise Welch's t-tests ($n = 10$ runs) confirm that on Lorenz 96, Ledoit-Wolf's advantage over all other methods is statistically significant ($p < 0.05$) at most ensemble sizes. On the cosmological data, however, most shrinkage methods do not differ significantly from Ledoit-Wolf ($p > 0.05$) at

Table 1. Time-averaged relative RMSE ($\times 10^{-2}$) for different methods and ensemble sizes. Best result per column in **bold**. †Scaled Shrinkage is a novel domain-agnostic adaptation proposed in this work (Section 3.2); all other methods follow Looijmans et al. [13].

Method	$N_e = 50$	$N_e = 60$	$N_e = 80$	$N_e = 100$
EnKF (standard)	0.35 ± 0.18	0.38 ± 0.56	0.08 ± 0.02	$\mathbf{0.06 \pm 0.01}$
EnKF-Cholesky	0.14 ± 0.01	0.13 ± 0.01	0.13 ± 0.01	0.13 ± 0.01
Identity Shrinkage	0.79 ± 0.62	1.53 ± 1.62	0.63 ± 0.31	0.81 ± 0.82
Scaled Shrinkage†	0.43 ± 0.20	0.88 ± 0.66	0.51 ± 0.24	0.59 ± 0.35
Eigenvalue Shrinkage	1.56 ± 0.59	2.00 ± 2.11	1.68 ± 0.93	1.30 ± 1.25
NERCOME	0.23 ± 0.09	0.29 ± 0.18	0.36 ± 0.15	0.42 ± 0.19
Ledoit-Wolf	$\mathbf{0.08 \pm 0.01}$	$\mathbf{0.08 \pm 0.01}$	$\mathbf{0.07 \pm 0.00}$	0.07 ± 0.01

Table 2. Relative Frobenius loss of precision matrix estimates on BOSS DR12 mock data ($d = 18$). Lower is better. Best result per column in **bold**.

Method	$n = 24$	$n = 30$	$n = 40$	$n = 50$
Sample (Hartlap)	1.15 ± 1.01	0.74 ± 0.44	0.47 ± 0.15	0.43 ± 0.16
Identity Shrinkage	1.18 ± 1.13	0.68 ± 0.43	$\mathbf{0.45 \pm 0.12}$	$\mathbf{0.41 \pm 0.14}$
Scaled Shrinkage†	0.90 ± 0.96	$\mathbf{0.60 \pm 0.36}$	0.51 ± 0.11	0.46 ± 0.13
Eigenvalue Shrinkage	1.20 ± 1.13	0.70 ± 0.43	0.46 ± 0.13	$\mathbf{0.41 \pm 0.14}$
Cosmological Shrinkage	1.19 ± 1.13	0.69 ± 0.43	0.46 ± 0.13	$\mathbf{0.41 \pm 0.14}$
NERCOME	$\mathbf{0.82 \pm 0.06}$	0.73 ± 0.08	0.63 ± 0.08	0.53 ± 0.08
Ledoit-Wolf	0.96 ± 0.01	0.95 ± 0.01	0.94 ± 0.01	0.93 ± 0.01

$N_e \geq 40$, confirming that the performance differences are modest; the consistent slight edge of Ledoit-Wolf across all four sizes is nonetheless notable.

The Part B results reveal a striking contrast with Part A. While Ledoit-Wolf was the worst performer in standalone precision estimation (Table 2), it achieves the best RMSE within the EnKF pipeline. The direct precision shrinkage methods—which excelled at standalone estimation—are competitive but do not lead. This gap between estimation quality and filtering performance is discussed further in Section 5.

5 Discussion

5.1 Numerical Behavior of Shrinkage Coefficients

Our diagnostic analysis reveals that the Bodnar et al. [3] closed-form coefficients α^* and β^* exhibit striking behavior in the Lorenz 96 setting. While α^* remains well-behaved at approximately $1-n/N_e$ across all ensemble sizes and assimilation steps, β^* is severely miscalibrated: its unclamped values reach 10^8–10^9 for the identity target and 10^3–10^5 for the scaled target, requiring clamping to $[0, 1]$

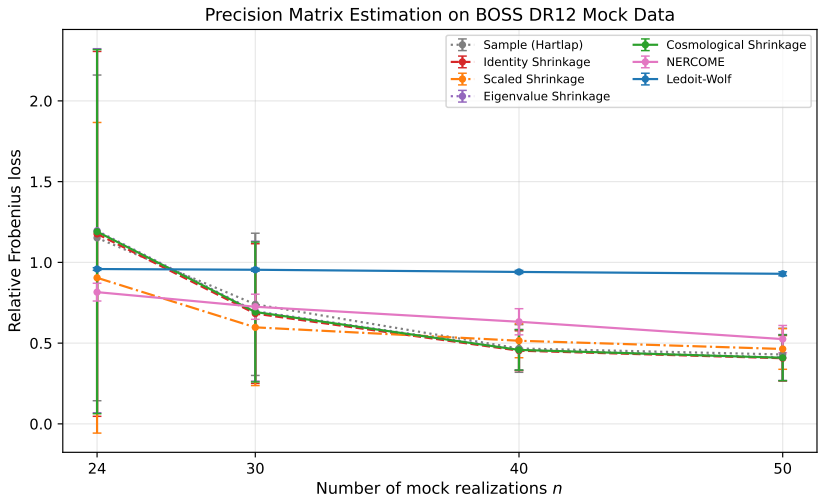

Fig. 5. Relative Frobenius loss of precision matrix estimates vs. subsample size on BOSS DR12 mock data ($d = 18$). Direct precision shrinkage methods (solid lines) achieve the lowest loss at $n \geq 40$, while Ledoit-Wolf (blue) remains flat near 0.93. Error bars show ± 1 standard deviation over 100 random draws.

at 100% of assimilation steps. For the eigenvalue target, β^* is effectively zero ($\sim 10^{-4}$), meaning no shrinkage toward the target occurs.

This behavior stems from the mismatch between the targets and the true precision structure of the Lorenz 96 system. The identity target $\boldsymbol{\Pi}_0 = \mathbf{I}$ is far from the true precision matrix, which has significant off-diagonal structure reflecting the spatial coupling in the Lorenz 96 dynamics. The Bodnar formula for β^* (Eq. 16) amplifies this mismatch through the $\mathrm{tr}(\hat{\boldsymbol{\Pi}}_S \boldsymbol{\Pi}_0)/\|\boldsymbol{\Pi}_0\|_F^2$ term, which becomes very large when the sample precision has large diagonal entries relative to the target norm.

The practical consequence is that the clamped estimator reduces to approximately $\boldsymbol{\Pi}_{\mathrm{LS}} \approx \alpha^* \hat{\boldsymbol{\Pi}}_S + (1 - \alpha^*)\boldsymbol{\Pi}_0$ for the identity and scaled targets, or $\boldsymbol{\Pi}_{\mathrm{LS}} \approx \alpha^* \hat{\boldsymbol{\Pi}}_S$ for the eigenvalue target. The resulting precision matrices have condition numbers of 10^7–10^{11}, leading to numerically unstable analysis updates that are highly sensitive to the random seed.

5.2 Effect of Shrinkage Target Choice

The experimental results reveal a clear distinction between covariance-domain and precision-domain shrinkage. The Ledoit-Wolf estimator, which operates in covariance space with an analytically optimal shrinkage intensity, achieves the lowest and most stable RMSE across all ensemble sizes. Its success reflects the fact that the Ledoit-Wolf framework is specifically designed for the regime where

Table 3. Time-averaged relative RMSE ($\times 10^{-2}$) for the AR(1) cosmological model. Best result per column in **bold**. †Scaled Shrinkage is a novel adaptation proposed in this work.

Method	$N_e = 24$	$N_e = 30$	$N_e = 40$	$N_e = 50$
EnKF (standard)	2.93 ± 0.44	2.20 ± 0.34	2.16 ± 0.25	2.06 ± 0.39
EnKF-Cholesky	2.07 ± 0.30	1.76 ± 0.20	1.86 ± 0.27	1.86 ± 0.37
Identity Shrinkage	2.03 ± 0.27	1.74 ± 0.23	1.86 ± 0.21	1.84 ± 0.34
Scaled Shrinkage†	2.04 ± 0.25	1.73 ± 0.20	1.89 ± 0.27	1.85 ± 0.33
Eigenvalue Shrinkage	2.39 ± 0.28	2.00 ± 0.26	2.01 ± 0.23	1.92 ± 0.35
Cosmological Shrinkage	2.25 ± 0.25	2.00 ± 0.28	2.05 ± 0.27	1.97 ± 0.36
NERCOME	1.93 ± 0.23	1.79 ± 0.21	1.88 ± 0.28	1.90 ± 0.33
Ledoit-Wolf	$\mathbf{1.94 \pm 0.27}$	$\mathbf{1.71 \pm 0.22}$	$\mathbf{1.84 \pm 0.22}$	$\mathbf{1.83 \pm 0.34}$

N_e is comparable to or larger than n, and its shrinkage intensity adapts automatically to the data.

In contrast, the direct precision shrinkage methods show erratic, non-monotonic behavior as N_e varies: RMSE can increase substantially at certain ensemble sizes before decreasing again. This instability is not a code error but a consequence of the Bodnar et al. formula producing coefficients outside their valid range, which the clamping mechanism cannot fully remedy. The methods were originally developed and validated for cosmological precision matrix estimation [13], where the targets (particularly the cosmological target $\boldsymbol{\Pi}_0^{(2)}$ based on the power spectrum) are well-matched to the true precision structure. When applied to a dynamical system with different covariance characteristics, the target mismatch degrades performance.

The modified Cholesky approach, by contrast, achieves consistently stable (though not optimal) performance across all ensemble sizes, making it a reliable baseline for precision matrix estimation in data assimilation.

The novel scaled identity target proposed in this work occupies a middle ground. On Lorenz 96, it outperforms the other direct precision shrinkage variants (Table 1), likely because scaling by empirical variances provides a better-conditioned target than the identity. On cosmological data, it achieves the lowest mean Frobenius loss at $n = 30$ (Table 2) but is slightly outperformed by the Bodnar methods at larger sample sizes. These results confirm its intended role as an exploratory, domain-agnostic alternative: it offers modest improvements over the identity target without requiring external prior knowledge, but cannot match domain-specific targets when such knowledge is available.

NERCOME exhibits a counterintuitive trend on Lorenz 96: its RMSE increases with ensemble size (0.23 at $N_e = 50$ to 0.42 at $N_e = 100$). This likely reflects the interaction between NERCOME's bootstrap resampling (200 draws per assimilation cycle) and the sequential nature of the EnKF. At small N_e, the sample covariance is noisy and NERCOME's resampling provides genuine regularization; at larger N_e, the sample covariance is already well-conditioned, and the bootstrap variance compounds over 100 sequential updates, degrading

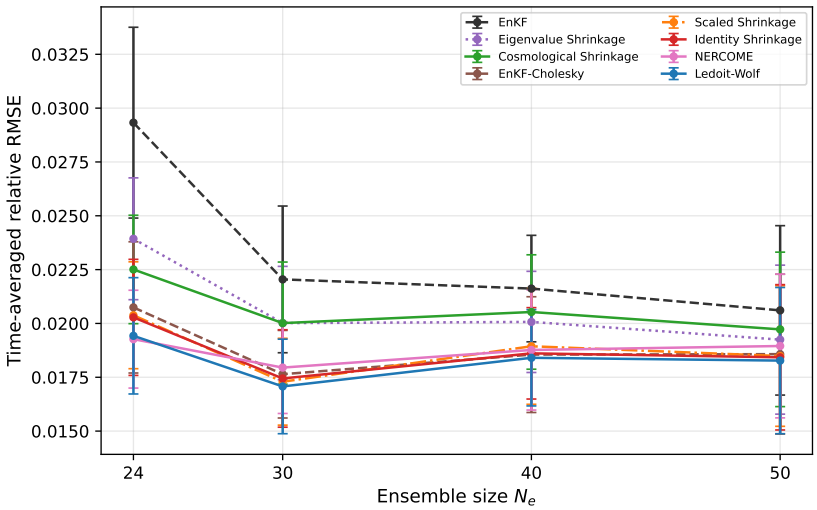

Fig. 6. Time-averaged relative RMSE vs. ensemble size for the AR(1) cosmological model. Unlike the Lorenz 96 case (Figure 3), all methods show stable behavior. Ledoit-Wolf (blue) achieves the lowest RMSE, while direct precision shrinkage methods are competitive. Error bars show ±1 standard deviation over 10 runs.

performance. This effect is absent on the cosmological data (Table 3), where NERCOME's performance is stable across ensemble sizes. NERCOME also carries a substantial computational cost: on the AR(1) model ($d = 18$, $N_e = 30$), it requires ∼27 seconds per 100 assimilation cycles compared to ∼0.2 seconds for the other shrinkage methods and ∼0.14 seconds for the standard EnKF. On Lorenz 96 ($n = 40$, $N_e = 80$), NERCOME takes ∼215 seconds versus ∼26 seconds for all other methods, an 8× overhead dominated by the bootstrap resampling.

5.3 Regime Dependence on N_e/n

The ratio of ensemble size to state dimension, N_e/n, plays a dual role. For the standard EnKF, larger ensembles monotonically improve performance, as expected from sampling theory. For the Ledoit-Wolf estimator, performance is nearly constant across the tested range ($N_e/n \in [1.25, 2.5]$), reflecting the effectiveness of its adaptive shrinkage.

For the direct precision shrinkage methods, no clear monotonic trend emerges on Lorenz 96. This contrasts with the cosmological setting, where these methods show stable improvement with increasing N_e/n (Table 3). The difference arises because in cosmological applications, the diagonal targets capture the dominant structure of the true precision matrix (which is approximately diagonal for independent multipole modes), whereas the Lorenz 96 system has dense off-diagonal

precision structure that none of the generic targets approximate well. Our dual-domain experiments (Section 4.7) confirm this explanation directly.

This finding has practical implications: the direct precision shrinkage framework of Bodnar et al. is most effective when domain-specific targets closely approximate the true precision structure. For general data assimilation problems where such prior knowledge is unavailable, covariance-domain approaches like Ledoit-Wolf may be preferable.

5.4 Educational Value

The integration of shrinkage estimators into TEDA serves several educational goals. First, it provides students with concrete implementations of methods that are otherwise encountered only in theoretical papers. The modular design of TEDA allows students to compare the effects of different precision matrix estimators on assimilation quality by changing a single parameter, keeping the dynamical model, ensemble configuration, and observation network fixed. This controlled comparison helps build intuition about the role of covariance estimation in data assimilation.

Second, the code structure mirrors the mathematical exposition: each algorithm in this paper corresponds to a class method in TEDA, with variable names matching the notation used here. This tight correspondence between theory and implementation reduces the cognitive gap that students often face when transitioning from textbook equations to working code.

Third, the contrast between methods that work well in theory but struggle in practice (direct precision shrinkage on Lorenz 96) and methods that are robustly effective (Ledoit-Wolf) provides a valuable lesson about the importance of assumptions and domain knowledge in applied mathematics.

5.5 Estimation Quality vs. Filtering Performance

The cosmological experiments reveal a surprising disconnect between precision matrix estimation quality and EnKF filtering performance. In Part A (Table 2), the direct precision shrinkage methods achieve a Frobenius loss of ≈ 0.41 at $n = 50$, far superior to Ledoit-Wolf's 0.93. Yet in Part B (Table 3), Ledoit-Wolf achieves the lowest RMSE in the EnKF pipeline, while the direct precision methods are only competitive.

This discrepancy arises because the EnKF applies the precision estimator at every assimilation cycle, compounding small differences across 100 sequential updates. Ledoit-Wolf's key advantage is *consistency*: its standard deviation across draws is ± 0.01 in Part A, an order of magnitude lower than the direct precision methods (± 0.13–0.14). In a sequential filtering context, an estimator that is slightly biased but highly stable can outperform one that is more accurate on average but more variable from step to step. This suggests that variance reduction, not just bias reduction, is critical for precision matrix estimation within data assimilation pipelines.

5.6 Domain Dependence: Lorenz 96 vs. Cosmological Data

Comparing the Lorenz 96 results (Table 1) with the cosmological results (Tables 2 and 3) confirms the central role of target-truth alignment. On Lorenz 96, where the true precision matrix has dense off-diagonal structure, the diagonal targets produce erratic RMSE with high variance (0.43–2.00×10^{-2}). On the cosmological data, where the precision is near-diagonal, the same methods achieve stable, competitive performance (1.73–2.39×10^{-2}) with substantially lower variance.

This dual-domain comparison validates the hypothesis that the poor Lorenz 96 performance of direct precision shrinkage is not a failure of the methods themselves, but a consequence of applying them outside their design regime. The Bodnar et al. framework is effective when the target captures the dominant structure of the true precision matrix.

5.7 Limitations

Several limitations should be noted. First, our current implementation does not combine shrinkage with spatial localization [6], which is another widely used technique for regularizing covariance matrices in data assimilation. The interaction between shrinkage and localization is an active area of research [15] and represents a natural extension of this work.

Second, the scaled identity target uses an ad-hoc scaling factor of 2.0, chosen as an analogy to the cosmological $2/N_\ell$ factor but without theoretical justification for the Lorenz 96 setting. Optimizing this parameter or deriving it from the data could improve performance.

Third, the AR(1) model used in Part B is a simplified dynamical system that preserves the covariance structure of the mock catalogs but lacks the complexity of realistic cosmological simulations. Nevertheless, it provides a controlled test of the TEDA pipeline with cosmological-like data.

6 Conclusions

We have presented the implementation of six shrinkage estimators within the TEDA framework: three direct precision shrinkage variants from Looijmans et al. [13] (identity, eigenvalue-based, and cosmological targets), one novel scaled identity adaptation proposed in this work, the NERCOME estimator of Joachimi [11], and the Ledoit-Wolf covariance shrinkage estimator [12]. These methods address the fundamental challenge of precision matrix estimation when the ensemble size is limited relative to the state dimension.

Our dual-domain experimental validation—using both the Lorenz 96 chaotic dynamical system and BOSS DR12 cosmological mock catalog data—reveals three key findings:

1. **Target-truth alignment is critical.** The direct precision shrinkage methods of Bodnar et al. [3] perform poorly on Lorenz 96 (dense off-diagonal

precision) but achieve excellent standalone estimation on cosmological data (near-diagonal precision), confirming the findings of Looijmans et al. [13] and demonstrating that these methods work as designed when the target matches the data structure.

2. **Estimation quality does not guarantee filtering performance.** On cosmological data, the Bodnar methods achieve the best standalone precision matrix estimates (Frobenius loss ≈ 0.41 vs. Ledoit-Wolf's 0.93), yet Ledoit-Wolf achieves the best RMSE within the EnKF pipeline. The stability of the estimator across sequential assimilation cycles matters as much as per-step accuracy.
3. **Ledoit-Wolf is robustly effective across domains.** It achieves the best or near-best EnKF performance in both the Lorenz 96 and cosmological settings, making it the safest default choice when domain-specific prior knowledge is unavailable.

The integration of these methods into TEDA maintains the framework's educational philosophy: each estimator is implemented as a self-contained class with clear mathematical documentation, enabling students and researchers to explore the effects of covariance regularization on data assimilation through controlled experiments. The contrast between methods that excel at standalone estimation but struggle in sequential filtering provides valuable pedagogical insight into the role of stability and variance in applied estimation theory.

Future work includes combining shrinkage with spatial localization methods, extending the implementation to support the quasi-geostrophic model for higher-dimensional experiments, and developing adaptive strategies that select shrinkage targets based on the observed covariance structure.

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